

3

In this chapter, we will learn about

- analysing circuits involving short circuit, potential divider and resistor network
- the internal resistance of ammeters, voltmeters and power sources

Circuits in Practice

3.1 Analysing circuit

1 Short circuit

DSE goal

Solve problems involving simple circuits.

DSE exam

12(1A)Q27, 13(1A)Q32, 14(1A)Q25, 15(1A)Q25, 19(1A)Q25, 22(1A)Q26

Flipped classroom

Watch a video and answer the questions.

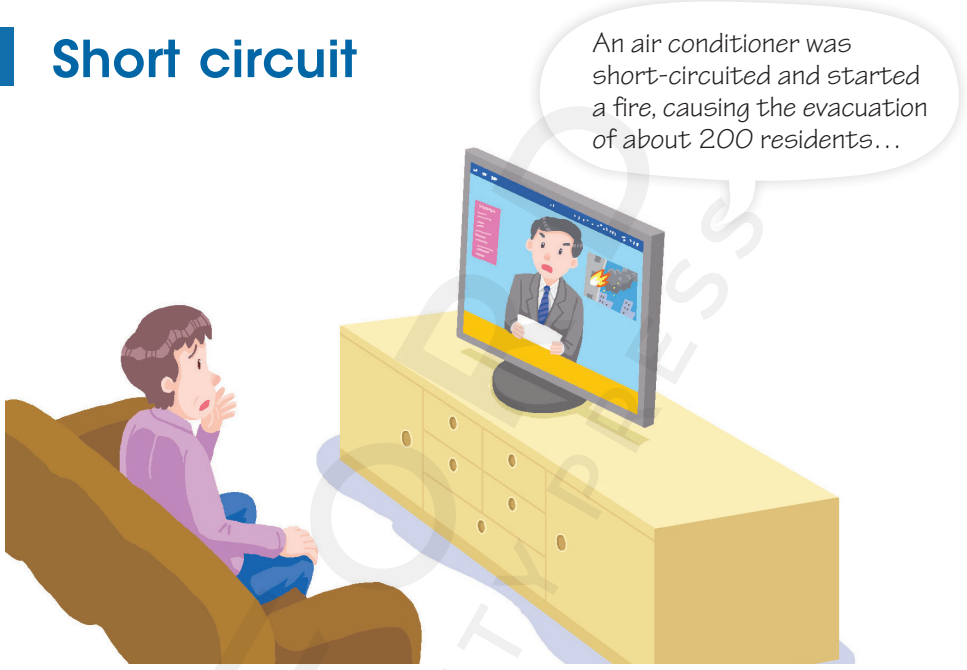


Fig 3.1a A short circuit may cause a fire.

We often hear people speak of a **short circuit**; however, what is it actually?

Consider Figure 3.1b. The bulb in the circuit lights up. In Figure 3.1c, a connecting wire is added across the bulb, forming a branch with almost zero resistance and in parallel to the bulb. As a result:

- the bulb goes out since nearly all the current flows through the new branch, not the bulb;
- the current drawn from the battery becomes larger since the total resistance of the circuit becomes lower.

We say that a short circuit is formed and the bulb is short-circuited (or shorted).

Recall (Ch 2 p.89) that when the bulb and the wire are parallel,

$$I_B = \frac{R_W}{R_W + R_B} \times I$$
$$\approx \frac{(0)}{(0) + R_B} \times I$$
$$= 0$$

R_{eq} of bulb and wire

$$= \frac{R_W R_B}{R_W + R_B} \approx \frac{(0) \times R_B}{(0) + R_B} = 0$$

Total resistance of circuit

$$= R + R_{eq} \approx R + (0) = R$$

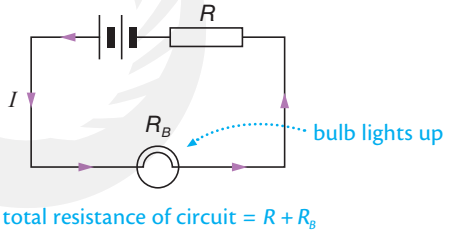


Fig 3.1b A bulb connected in series to a battery.

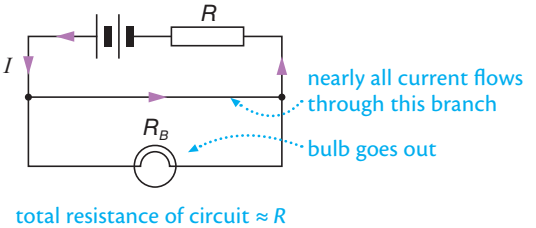


Fig 3.1c A bulb does not light up when it is shorted.

If a circuit component is shorted, the current through the circuit will increase. The circuit may become very hot and cause a fire.

Even a battery can cause a fire when shorted.

Video

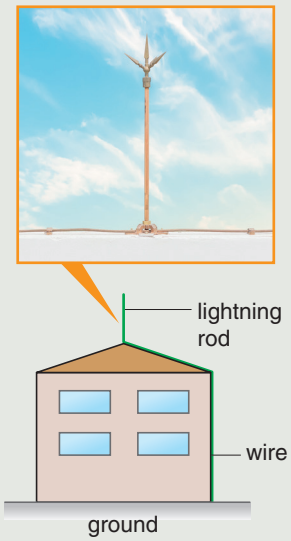
→ This video shows how a 9-V battery causes a fire when shorted.

short circuit 短路

Know more

Lightning rod

A lightning rod is installed at the top of a building and is connected to the ground with a wire. If a lightning strikes the lightning rod, the wire shorts the building, so the lethal current flows to the ground via the wire instead of the building.

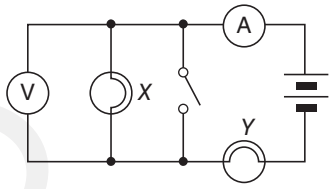


Google Forms

Microsoft Forms

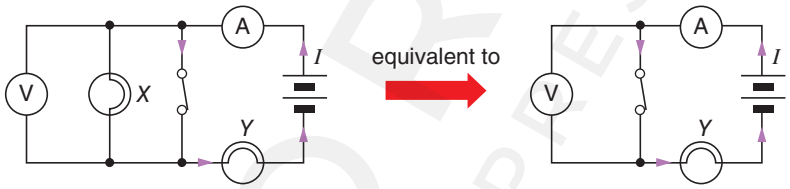
Example 1 Short circuit

In the circuit shown, how do the readings of the ammeter and the voltmeter change when the switch is closed? Explain.



Solution

When the switch is closed, bulb *X* is shorted.



The total resistance of the circuit decreases and the current through the circuit increases. Therefore, the ammeter reading increases.

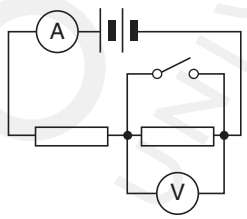
The voltmeter is now connected across a branch of almost zero resistance, so its reading decreases.

Quick check 1

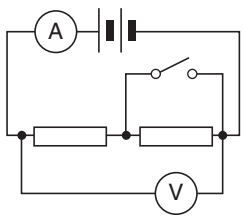
E-version in OUP Exercise Platform is also available.

1 How do the readings of the ammeter and voltmeter change when the switch is closed?

(a)



(b)



Learn by practice 1

1 A circuit is connected as shown in Figure a.

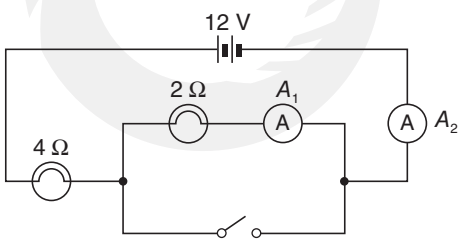


Fig a

What are the ammeter readings before and after closing the switch?

2 In Figure b, the switch is closed initially.

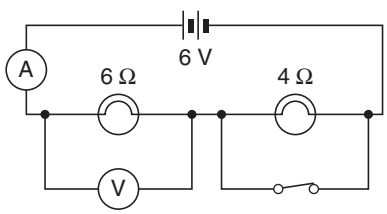


Fig b

How do the readings of the ammeter and the voltmeter change once the switch is opened? Explain.

DSE goal

Know the concept of potential divider.

DSE exam

13(1A)Q31, 13(1B)Q10(c), 22(1B)Q8(a)

Know more

Rheostat as potential divider

A rheostat can be used as a potential divider.

▸ p.144

2 Potential divider

Consider the circuit below (Fig 3.1d). We learned in Chapter 2 (p.87):

$$V_1 = \frac{R_1}{R_1 + R_2} \times \varepsilon$$

$$V_2 = \frac{R_2}{R_1 + R_2} \times \varepsilon$$

$$V_1 : V_2 = R_1 : R_2$$

This means that the two resistors 'divide' the e.m.f. ε according to the ratio of their resistance $R_1 : R_2$. By selecting resistors of suitable resistance ratio, we can harvest any output voltage V_{out} smaller than ε (Fig 3.1e). This connection is called a **potential divider** (Fig 3.1f).

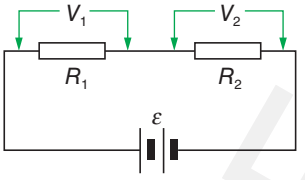


Fig 3.1d *Resistors R_1 and R_2 connected in series to a battery.*

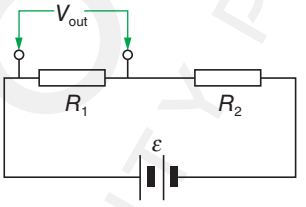


Fig 3.1e *The output voltage depends on the resistance ratio.*

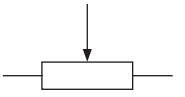


Fig 3.1f *Circuit symbol of potential divider.*

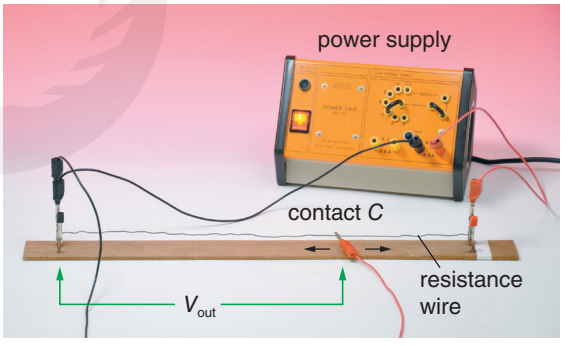
If $R_1 >> R_2$, $V_{out} \approx \varepsilon$ ▶

output **voltage** (V) ⋯▸ $V_{out} = \left(\frac{R_1}{R_1 + R_2}\right)\varepsilon$ ◂⋯ **e.m.f.** of source (V)

ratio of **resistance** (no unit)

Figure 3.1g shows a potential divider formed by a uniform resistance wire. V_{out} is determined by the position of contact C. Since the resistance R is directly proportional to the length l of the wire,

$$V_{out} = \left(\frac{R_1}{R_1 + R_2}\right)\varepsilon = \left(\frac{l_1}{l_1 + l_2}\right)\varepsilon$$



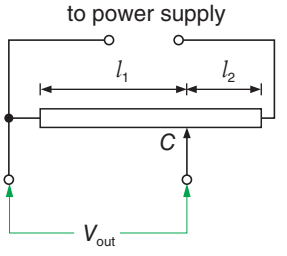
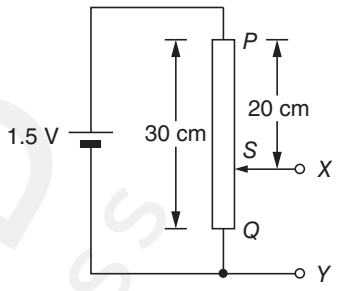


Fig 3.1g *Using a uniform resistance wire as a potential divider.*

potential divider 分壓器

Example 2 | Potential divider

A uniform resistance wire PQ is connected to a 1.5-V battery. It is 30 cm long and has a total resistance of 120 Ω. S is a sliding contact which can move along PQ. Initially PS is 20 cm long.



(a) Find the voltage V across X and Y.

(b) Then a light bulb is connected across X and Y. How will the brightness of the bulb change when S is moved towards Q?

Solution

(a) $V = \frac{SQ}{PQ} \times \varepsilon = \frac{30 - 20}{30} \times 1.5 = 0.5$ V

(b) When S is moved towards Q, the voltage across the bulb decreases. Therefore, the bulb becomes dimmer.

Caution

When the bulb is connected across X and Y, R_{eq} of the bulb and SQ is lower than the resistance of SQ, so the voltage across the bulb is lower than 0.5 V.

Try Exercise 3.1 Q5 (p.119) for some calculation.

Quick check 2

E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

(For Q1–2.) Three resistors are connected to a battery as shown (Fig a).

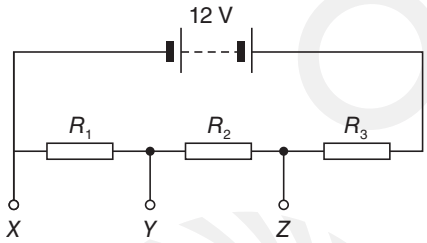


Fig a

1 Complete the table below.

	R_1 / Ω	R_2 / Ω	R_3 / Ω	V_{XY} / V	V_{XZ} / V
(a)	3	2	1		
(b)	100		200	2	

2 A bulb is the brightest when connected across

A X and Y.

B X and Z.

C Y and Z.

(For Q3–4.) A uniform resistance wire PQ is connected to a 6-V battery (Fig b). S is a sliding contact which can move along PQ.

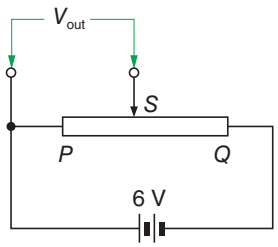


Fig b

3 Complete the table below.

	PQ / cm	PS / cm	V_{out} / V
(a)	100	25	
(b)		10	3

4 If S is moved towards Q, V_{out} will

A increase.

B remain unchanged.

C decrease.

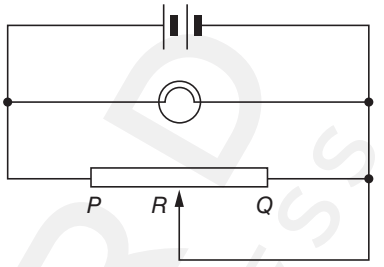
☆ HKDSE 2013 Paper 1A Q31

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Brightness and potential divider

A light bulb and a resistance wire *PQ* are connected to a battery as shown. A sliding contact *R* is moved along the resistance wire.



When *R* is moved towards end *P*, the bulb

A becomes dimmer.

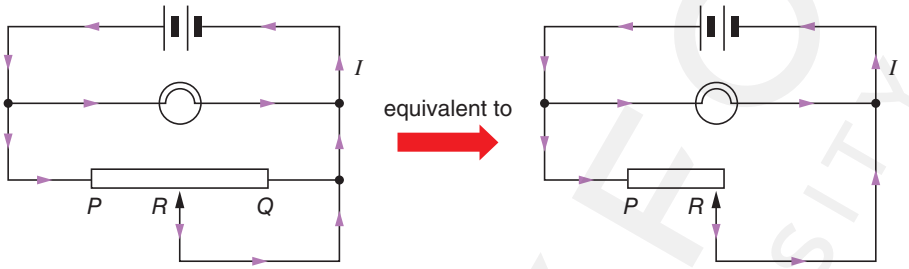
B becomes brighter.

C goes out when *R* reaches *P*.

D remains unchanged.

Solution

When *R* touches the resistance wire, the section *RQ* is shorted.



Since the bulb and *PR* are connected in parallel to the battery,
p.d. across bulb = p.d. across *PR* = e.m.f. of battery = constant

∴ The p.d. across the bulb, hence the brightness, does not change with the position of *R*.

∴ The answer is D.

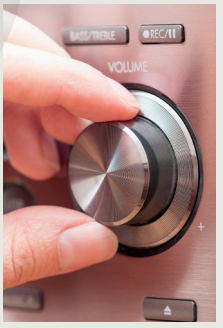
Common mistake ⚠
Students may not notice that the p.d. across the bulb is not affected by the change in resistance in the parallel branch. Even when *R* is moved to *P* and forms a short circuit, the p.d. across the bulb still remains unchanged. However, the current through the short circuit is very large and the battery goes flat very quickly.

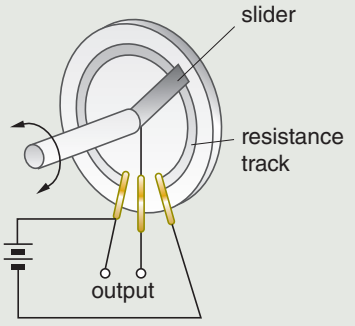
▶ Revision 3 Q9 (p.137)

Know more

Application of potential divider

Potential dividers can vary the output voltage and can be used in volume controls of audio equipment and in light dimmers.





DSE goal

Solve problems involving simple circuits.

DSE exam

12(1A)Q26, 16(1A)Q27, 17(1A)Q24, 21(1A)Q23, 21(1B)Q8, 23(1B)Q7(b)

Simulation

→ This simulation shows 6 circuits in which the resistance and/or type of each component can be changed. Ss will find out how this affects the current through and the voltage across each component.

→ Teaching notes and exercises are available on OUP web.

3 Resistor network

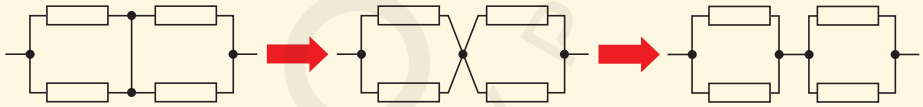
We learned in Chapter 2 (p.91) how to find the equivalent resistance when some resistors are connected in series and some in parallel. This skill can be applied to any number of resistors.

Sometimes, it may not be easy to identify which resistors are in series and which are in parallel. We may rearrange the network by using the following skill.

Skill *Problem-solving*

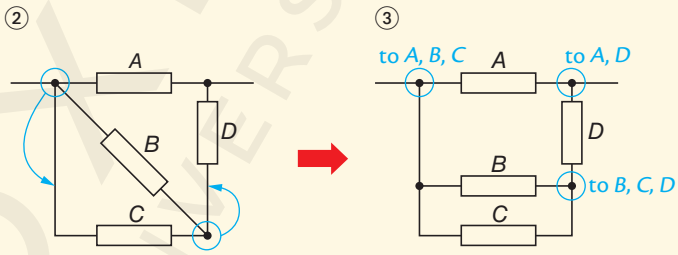
Redrawing a complicated network

① Wires can be lengthened or shortened. Two junctions connected by a wire can be joined as one, and vice versa.



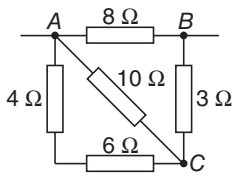
② Rearrange the wires to place the resistors either horizontally or vertically. You can move a junction along a wire.

③ Make sure that all the junctions connect to the same resistors as in the original circuit.



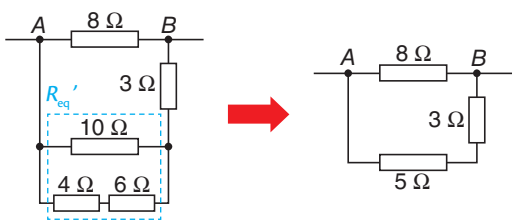
Example 3 Complicated network

Some resistors are connected as shown. Find the equivalent resistance R_{eq} of the network across A and B.



Solution

The network is equivalent to the one as shown.



$$R_{eq}' = \frac{10 \times (4 + 6)}{10 + (4 + 6)} = 5\ \Omega$$

$$R_{eq} = \frac{8 \times (5 + 3)}{8 + (5 + 3)} = 4\ \Omega$$

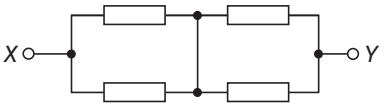
Google Forms Microsoft Forms

Quick check 3

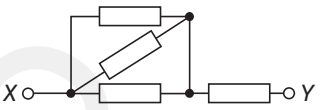
E-version in OUP Exercise Platform is also available.

1 Find the equivalent resistance across X and Y. Each resistor has a resistance of 3 Ω.

(a)

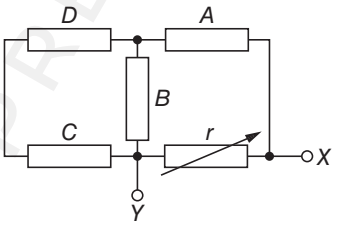


(b)



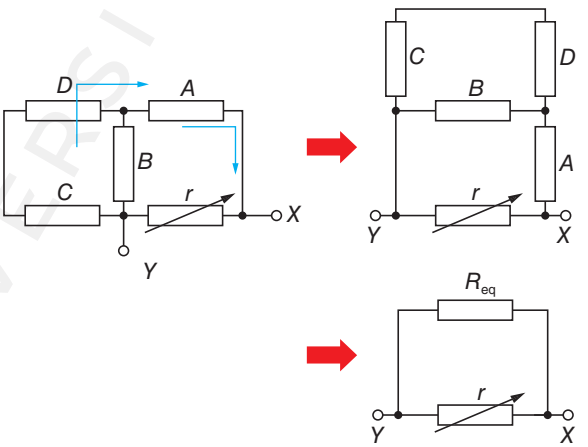
Example 4 Variable resistor and resistor network

In the figure, variable resistor r is connected with four identical resistors A, B, C and D. When r is set to 24 Ω, the equivalent resistance R_{XY} across X and Y is 15 Ω. Find r when R_{XY} is 8 Ω.



Solution

Resistors A to D can be replaced by a resistor of resistance R_{eq}.



When r = 24 Ω,

$$\frac{1}{24} + \frac{1}{R_{eq}} = \frac{1}{15}$$

$$R_{eq} = 40\ \Omega$$

When R_{XY} = 8 Ω,

$$\frac{1}{r} + \frac{1}{40} = \frac{1}{8}$$

$$r = 10\ \Omega$$

Tip
You need not calculate the resistance of each resistor.

Revision 3 Q15 (p.138)

Exercise 3.1

★**1** The bulbs are identical in Figure a.

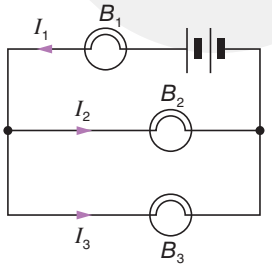


Fig a

Which of the following is/are correct?

(1) $I_2 = I_3$

(2) I_2 does not change if B_3 is shorted.

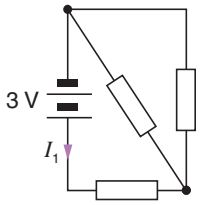
(3) B_2 becomes brighter if B_3 burns out.

A (1) only B (2) only

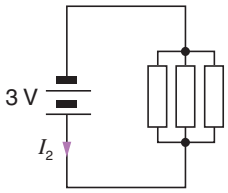
C (1) and (3) only D (2) and (3) only

★**2** Three identical resistors are connected to a 3-V battery in different ways.

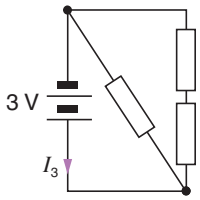
(1)



(2)



(3)



Which of the following is correct?

A $I_1 > I_2 > I_3$ B $I_1 > I_3 > I_2$

C $I_2 > I_3 > I_1$ D $I_2 > I_1 > I_3$

★**3** Three light bulbs are connected to a battery as shown (Fig b). The switch is closed initially. How will the brightness of the bulbs change when the switch is opened?

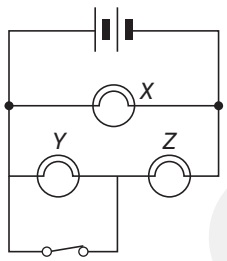


Fig b

	X	*Y*	*Z*
A	increases	decreases	increases
B	increases	increases	decreases
C	unchanged	decreases	increases
D	unchanged	increases	decreases

★**4** Consider the circuit shown in Figure c.

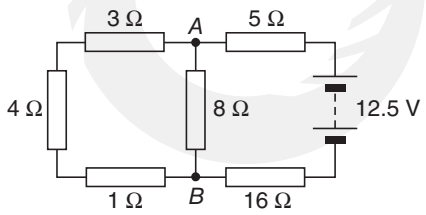


Fig c

(a) Find the equivalent resistance of the whole resistor network.

(b) Find the p.d. across *AB*.

(c) Find the current passing through the 4-Ω resistor, 8-Ω resistor, and 16-Ω resistor.

(d) How do the answers to (c) change if *AB* is shorted?

★**5** In the circuit shown in Figure d, *PQ* is a uniform wire 10 cm long with a total resistance of 50 Ω. The light bulb has a resistance of 30 Ω and is connected to *PQ* with a sliding contact *S*. Initially, *S* is 4 cm from *Q*.

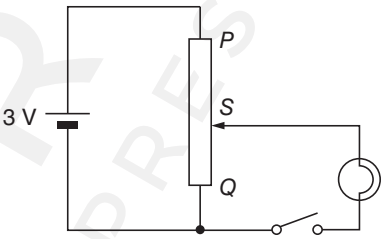


Fig d

(a) The switch is open initially.

(i) Find the voltage across *SQ*.

(ii) Find the resistance of wire segment *SQ*.

(b) Then the switch is closed.

(i) Find the equivalent resistance of the bulb and wire segment *SQ*.

(ii) Find the voltage across *SQ*.

(iii) Find the power consumed by the bulb.

(iv) How does the brightness of the bulb change when *S* is moved towards *P*?

(v) What is the maximum power consumed by the bulb?

★**6** The resistance of a photoresistor (Fig e) increases when the amount of light falling on it decreases. In Figure f, a switch is connected to a lighting system. It turns on the lights when the voltage across it is above a certain value. Complete the circuit in Figure f by connecting a fixed resistor and a photoresistor to the circuit in such a way that the circuit can turn on the lights automatically when the environment is dark. Explain the action of the circuit.

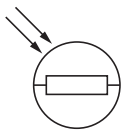


Fig e

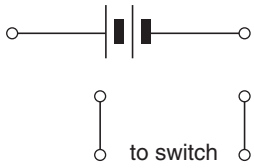


Fig f

3.2 Practical ammeters, voltmeters and batteries

DSE goal

Explain the effects of resistance of ammeters and voltmeters on measurements.

DSE exam

13(1B)Q10(b), 16(1B)Q7, 19(1A)Q24, 20(1A)Q23

1 Effects of ammeter and voltmeter

Recall Chapter 2.1–2.2. Do you know why a voltmeter can be connected directly to a battery while an ammeter cannot (Fig 3.2a)? This is related to their **internal resistance**.



Fig 3.2a *A voltmeter can be connected directly to a battery while an ammeter cannot.*

All circuit components, including ammeters and voltmeters, have their own resistance which is called the internal resistance. In this part, we will study how internal resistance affects a circuit.

a Internal resistance of ammeter

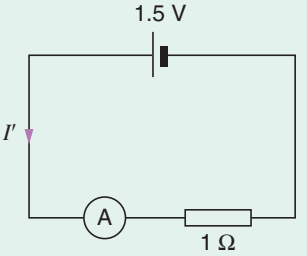
Video



→ This video demonstrates Expt 3a.

Experiment 3a / Internal resistance of ammeter

1 Connect the circuit as shown.



2 Record the ammeter reading I'.

3 Calculate the current I passing through the resistor if it is connected to the battery without the ammeter.

Discussion

Are I and I' the same? Why?

May ask Ss to add more ammeters and observe what happens.

In the experiment, I is the theoretical value of the current flowing through the circuit. I' is the measured value of the current when an ammeter is added to the circuit. The experiment shows that $I \neq I'$. This is because a practical ammeter has a non-zero internal resistance R_A.

internal resistance 內電阻

In Figure 3.2b, when an ammeter is added to the circuit, the total resistance of the circuit increases from R to $(R + R_A)$. This reduces the current passing through the circuit. If R_A is small, the ammeter will give a small effect on the circuit and I' will be close to I.

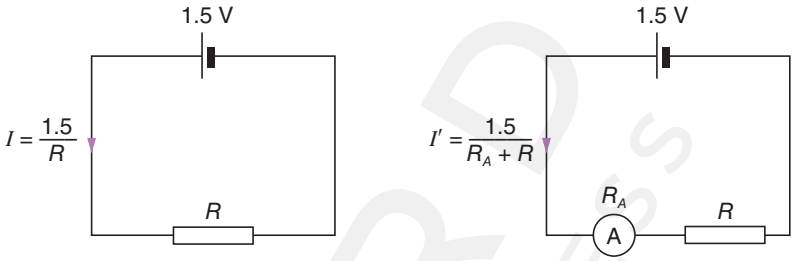


Fig 3.2b *Circuits with and without an ammeter.*

- **An ammeter reduces the current passing the circuit.**
- **A good ammeter has a very low internal resistance.**

An ammeter has a very low internal resistance. If it is connected across a battery directly, it will be damaged by the large current.

May tell Ss that the p.d. across the ammeters ≈ 0.1 V.

Table 3.2a shows the typical internal resistance of some ammeters.

Range of ammeter	Internal resistance / Ω
0–1 A	0.1
0–1 mA (milliammeter)	100
0–100 μA (microammeter)	1000

Table 3.2a *Typical internal resistance of some ammeters.*

Caution

The internal resistance of some ammeters is not actually very low. By 'very low', we mean in comparison to the resistance of the circuit.

An **ideal ammeter has zero resistance**. It has no effect on a circuit when connected in series.

Google Forms Microsoft Forms

Quick check 4

E-version in OUP Exercise Platform is also available.

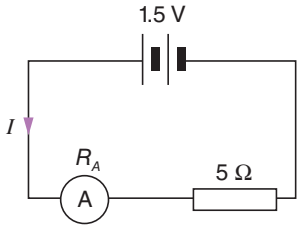
1 *True or false:* A good ammeter has a very low resistance. (T/F)

2 Consider the circuit shown. Find I when

(a) the ammeter is ideal;

(b) $R_A = 0.1\ \Omega$;

(c) $R_A = 1\ \Omega$.



b Internal resistance of voltmeter

Video

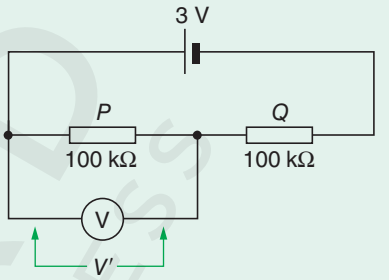
→ This video demonstrates Expt 3b.

Experiment 3b / Internal resistance of voltmeter

1 Connect the circuit as shown.

2 Record the voltmeter reading V'.

3 Calculate the voltage V across resistor P when the voltmeter is removed.



Discussion

Are V and V' the same? Why?

✎ May ask Ss to add more voltmeters and observe what happens.

In the experiment, V and V' are the theoretical value and measured value of the voltage across resistor P respectively. They are not equal because of the internal resistance R_V of the practical voltmeter (Fig 3.2c).

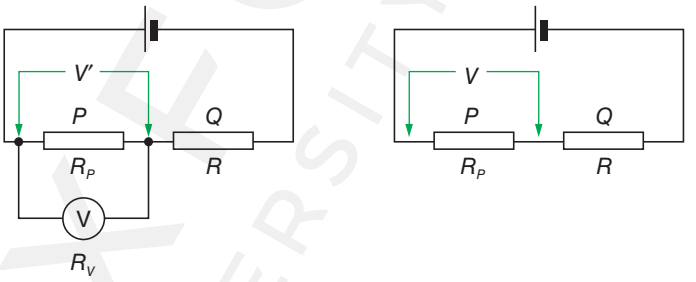


Fig 3.2c Circuits with and without a voltmeter.

Adding the voltmeter creates a branch for current to flow. The equivalent resistance of P and the voltmeter is lower than R_P, hence the voltage across P is reduced. If R_V is large, the current passing through the voltmeter will be small. As a result, the voltmeter will give a small effect on the circuit and V' will be close to V.

- **A voltmeter reduces the voltage across the component it measures.**
- **A good voltmeter has a very high internal resistance.**

No current passes through an ideal voltmeter. ▶

An **ideal voltmeter has infinite internal resistance**. Table 3.2b shows the typical internal resistance of some practical voltmeters.

Caution
By 'very high', we mean in comparison to the resistance of the component. ▶

Range of voltmeter / V	Internal resistance / Ω
0–0.1	10^3
0–1	10^4
0–10	10^5

Table 3.2b Typical internal resistance of some practical voltmeters.

Example 5 Effect of internal resistance of voltmeter

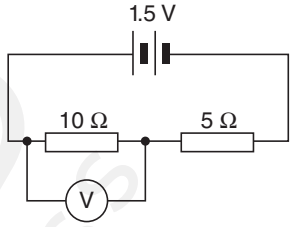
Consider the circuit shown.

Find the reading of the voltmeter if

(a) it is ideal;

(b) its internal resistance is 30 Ω;

(c) its internal resistance is 1000 Ω.



Solution

(a) Reading of voltmeter $= \frac{10}{10+5} \times 1.5 = 1$ V

(b) R_{eq} of 10-Ω resistor and voltmeter $= \frac{10 \times 30}{10+30} = 7.5$ Ω

Reading of voltmeter $= \frac{7.5}{7.5+5} \times 1.5 = 0.9$ V

% error $= \frac{1-0.9}{1} \times 100\%$
$= 10\%$

(c) R_{eq} of 10-Ω resistor and voltmeter $= \frac{10 \times 1000}{10+1000} = 9.90$ Ω

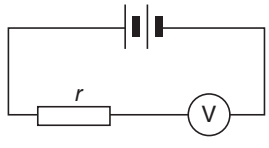
Reading of voltmeter $= \frac{9.90}{9.90+5} \times 1.5 = 0.997$ V

% error $= \frac{1-0.997}{1} \times 100\%$
$= 0.3\%$

An ideal voltmeter has no effect on a circuit when connected in parallel. However, if it is connected **in series**, an **open circuit** will be formed (Fig 3.2d).

r is negligible when compared with the internal resistance of an ideal voltmeter.

Note that this circuit is like connecting the voltmeter directly to the battery, so the voltmeter reading is **not zero** but equal to the e.m.f. of the battery.



circuit becomes open and no current flows

Fig 3.2d *An ideal voltmeter connected in series.*

Quick check 5

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:* A good voltmeter has a very low resistance. (T/F)

2 The voltmeter in the circuit is ideal (Fig a). Find its reading.

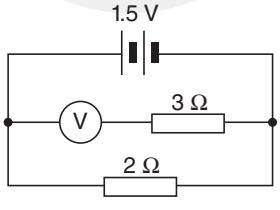


Fig a

3 Consider the circuit shown in Figure b. Find the voltmeter reading when

(a) the voltmeter is ideal;

(b) $R_V = 1$ Ω;

(c) $R_V = 1$ kΩ.

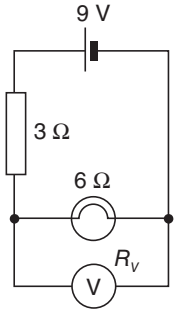


Fig b

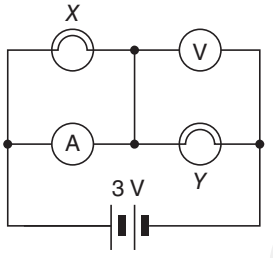
C Using ideal ammeter and voltmeter

☆ HKDSE 2019 Paper 1A Q24, 2020 Paper 1A Q23

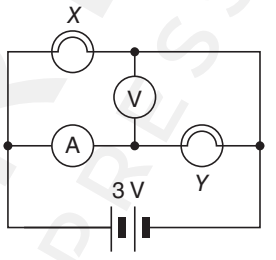
Example 6 Network with ideal ammeter and voltmeter

Light bulbs X and Y have the same resistance of 2 Ω. What are the readings of the ammeter and voltmeter in the following circuits? The ammeter and voltmeter are ideal.

(a)

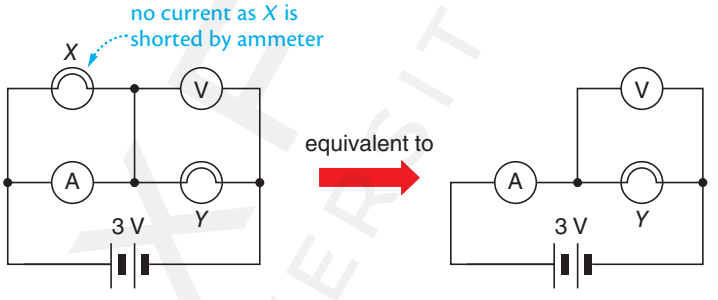


(b)



Solution

(a)

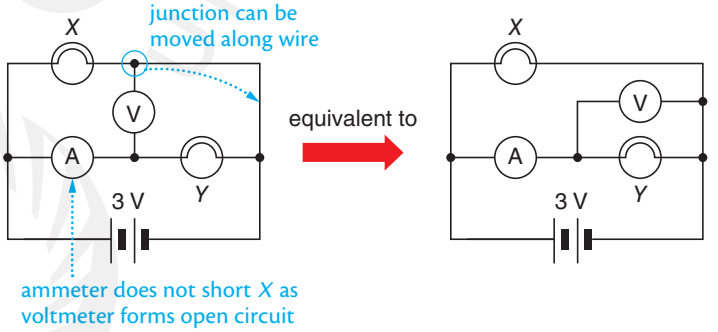


Tip
Y is the only load in the circuit.

Voltmeter reading = ε = 3 V

Ammeter reading = $\frac{V_Y}{R_Y} = \frac{3}{2}$ = 1.5 A

(b)



Tip
X and *Y* are connected in parallel to the battery.

Voltmeter reading = ε = 3 V

Ammeter reading = $\frac{V_Y}{R_Y} = \frac{3}{2}$ = 1.5 A

Revision 3 Q4 (p.136)

Think about...

What is the current through the battery in (a) and (b)?

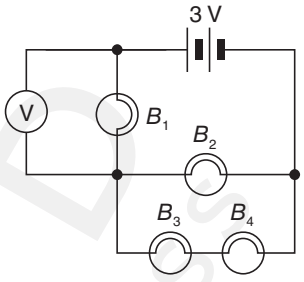
☆ HKDSE 2013 Paper 1A Q32, 2014 Paper 1A Q25

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Shorted or burns out?

Four light bulbs are connected to a battery as shown. The voltmeter is ideal and its reading is 3 V. Which of the following is/are possible?



(1) B_1 is shorted.

(2) B_1 burns out.

(3) B_2 is shorted.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

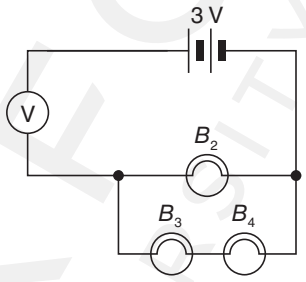
Solution

If B_1 is shorted, the voltmeter is connected across a part of zero resistance, so its reading is zero.

∴ (1) is incorrect.

Common mistake ⚠
Students may not fully understand the meaning of short circuit and open circuit.

If B_1 burns out, the circuit becomes open. Since the voltmeter is connected to the two terminals of the battery, it measures the e.m.f. of the battery, i.e. 3 V.



The resistance of B_2, B_3 and B_4 is negligible when compared with the internal resistance of the ideal voltmeter.

∴ (2) is correct.

If B_2 is shorted, B_3 and B_4 are also shorted.

B_1 becomes the only load in the circuit, so p.d. across it = 3 V.

∴ (3) is correct.

∴ The answer is D.

Revision 3 Q10 (p.137)

d Using practical ammeter and voltmeter

Cross-link
See Ch 2.3 (p.74).

Recall that the voltmeter-ammeter method can be used to find the resistance r of a circuit component. The two meters can be connected in two ways (Fig 3.2e). If the meters are ideal, both connections will give the theoretical values. If the meters are not ideal, the suitable connection will be chosen according to the value of r.

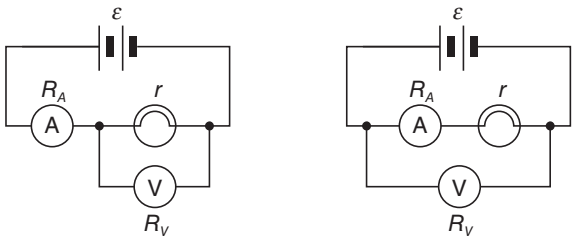
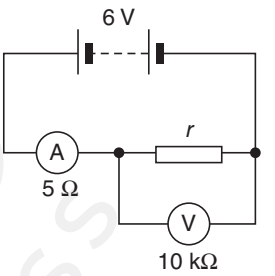


Fig 3.2e *Two connections of voltmeter-ammeter method.*

Example 7 Voltmeter-ammeter method

In the circuit shown, an ammeter of resistance 5 Ω and a voltmeter of resistance 10 kΩ are used to determine the resistance r of a resistor.



(a) What does the ammeter measure?

(b) What does the voltmeter measure?

(c) Find the percentage error in the value of r obtained using this circuit if

(i) $r = 2\ \Omega$;

(ii) $r = 2\ \text{k}\Omega$.

Solution

(a) It measures the total current through r and the voltmeter.

(b) It measures the voltage across r.

(c) (i) R_{eq} of voltmeter and $r = \dfrac{R_V r}{R_V + r} = \dfrac{10\,000 \times 2}{10\,000 + 2} = 1.9996\ \Omega$

Total resistance of circuit $= R_A + R_{eq} = 5 + 1.9996 = 6.9996\ \Omega$

Ammeter reading $I = \dfrac{\varepsilon}{R_A + R_{eq}} = \dfrac{6}{6.9996} = 0.857\,19\ \text{A}$

Voltmeter reading $V = \dfrac{R_{eq}}{R_A + R_{eq}} \times \varepsilon = \dfrac{1.9996}{6.9996} \times 6 = 1.7140\ \text{V}$

Measured value of $r = \dfrac{V}{I} = \dfrac{1.7140}{0.857\,19} = 1.9996\ \Omega$

Percentage error $= \dfrac{1.9996 - 2}{2} \times 100\%$

$= -0.0222\%$

(ii) R_{eq} of voltmeter and $r = \dfrac{R_V r}{R_V + r} = \dfrac{10\,000 \times 2000}{10\,000 + 2000} = 1666.7\ \Omega$

Total resistance of circuit $= R_A + R_{eq} = 5 + 1666.7 = 1671.7\ \Omega$

Ammeter reading $= \dfrac{\varepsilon}{R_A + R_{eq}} = \dfrac{6}{1671.7} = 3.5892 \times 10^{-3}\ \text{A}$

Voltmeter reading $= \dfrac{R_{eq}}{R_A + R_{eq}} \times \varepsilon = \dfrac{1666.7}{1671.7} \times 6 = 5.9821\ \text{V}$

Measured value of $r = \dfrac{V}{I} = \dfrac{5.9821}{3.5892 \times 10^{-3}} = 1666.7\ \Omega$

Percentage error $= \dfrac{1666.7 - 2000}{2000} \times 100\% = -16.7\%$

Exercise 3.2 Q5 (p.133)

Caution
The ammeter reading is larger than the current through r.

Since the deviation is small, 5 sig. fig. are used.

Caution
The voltmeter reading is **not** Ir.

Tip
Measured value of r

$= \dfrac{V}{I}$

$= \dfrac{\frac{R_{eq}}{R_A + R_{eq}} \times \varepsilon}{\frac{\varepsilon}{R_A + R_{eq}}}$

$= R_{eq}$

We can see that the connection in Example 7 is suitable for measuring a **low resistance**. When R_V is much larger than r, the current through the voltmeter, hence the error in the ammeter reading, is small, so the error in the measured value of r is small.

Figure 3.2f shows the connection suitable for measuring a **high resistance**. The ammeter measures the current through r. The voltmeter measures the total voltage across r and the ammeter. When R_A is much smaller than r, the voltage across the ammeter, hence the error in the voltmeter reading, is small, so the error in the measured value of r is small. Mathematically,

Voltmeter reading > voltage across r

$$\text{measured value of } r = \frac{\text{voltmeter reading}}{\text{ammeter reading}} = \frac{IR_A + Ir}{I} = R_A + r$$

$$\text{error in value of } r = \text{measured value} - \text{actual value} = (R_A + r) - r = R_A$$

$$\text{percentage error in value of } r = \frac{R_A}{r} \times 100\%$$

Therefore, when $r >> R_A$, the percentage error will be small.

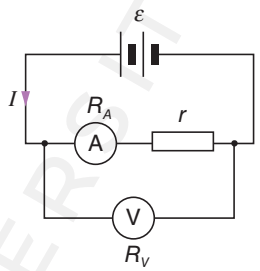


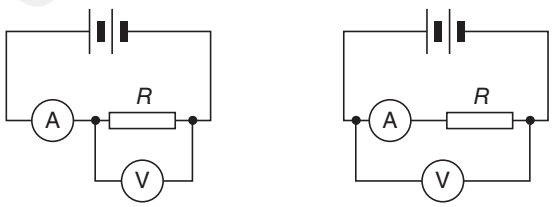
Fig 3.2f *Connection suitable for measuring high resistance.*

Google Forms Microsoft Forms

Quick check 6

E-version in OUP Exercise Platform is also available.

1 Practical ammeter and voltmeter are used to measure resistance R.



circuit 1 **circuit 2**

(a) Which circuit is more suitable for measuring R if R is lower than 10 Ω?

(b) In which circuit does the voltmeter show the voltage across R?

(c) In which circuit does the ammeter show the current flowing through R?

(d) In which circuit is the measured value of R greater than the actual value?

DSE goal

Compare e.m.f. of source and terminal voltage across the source experimentally and relate the difference to internal resistance.

DSE exam

12(1A)Q28, 14(1A)Q24, 15(1A)Q26, 19(1B)Q7

Video

→ This video demonstrates Expt 3c.

→ This is a Video Quiz about this experiment.

2 Internal resistance of power source

Experiment 3c / Internal resistance of battery

1 Connect a voltmeter across a 6-V battery (Fig a). Record the voltmeter reading V_1.

2 Connect a lamp to the battery (Fig b). Record the voltmeter reading V_2.

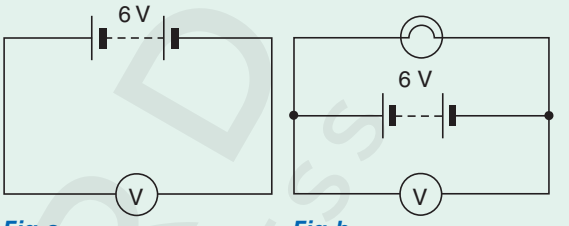


Fig a *Fig b*

Discussion

Are V_1 and V_2 the same? Why?

In step 1 of the experiment, the voltmeter reading, i.e. the voltage across the battery, is about the same as the e.m.f. After connecting the lamp, the voltage across the battery drops (Fig 3.2g). This actual output voltage of the battery is called **terminal voltage**.

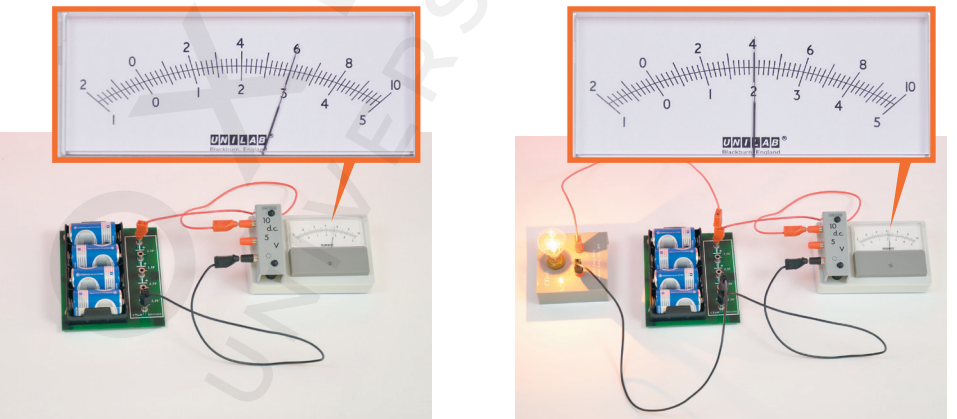
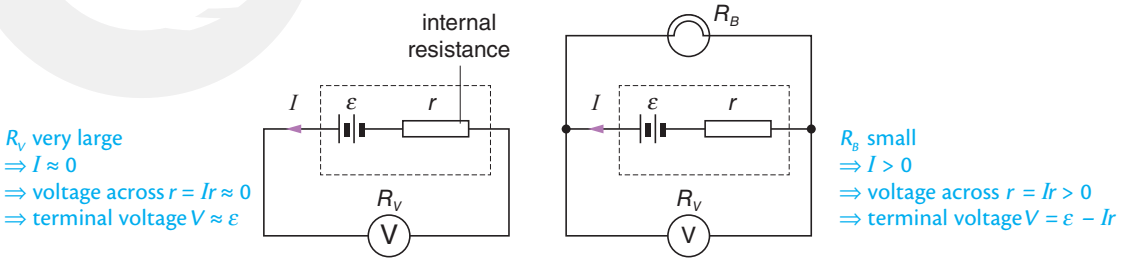


Fig 3.2g *The voltmeter reading drops after the lamp is connected.*

The drop in voltage is due to the internal resistance of the battery. We can regard the battery as connected in series to a resistor r which causes the internal resistance (Fig 3.2h). When current flows in the circuit, r shares some voltage, so the terminal voltage is lower than the e.m.f. ε.



R_V very large
$\Rightarrow I \approx 0$
$\Rightarrow$ voltage across $r = Ir \approx 0$
$\Rightarrow$ terminal voltage $V \approx \varepsilon$

R_B small
$\Rightarrow I > 0$
$\Rightarrow$ voltage across $r = Ir > 0$
$\Rightarrow$ terminal voltage $V = \varepsilon - Ir$

Fig 3.2h *Terminal voltage and internal resistance of battery.*

terminal voltage 端電壓

Tip
V decreases if *I* or *r* increases.

$$\varepsilon = V + Ir$$
e.m.f. (V); terminal voltage (V); internal resistance (Ω); current through battery (A)

Know more

Battery at low temperature

A smartphone may run out of battery easily in a very cold environment. This is because the internal resistance of the battery increases when the temperature drops. This makes the terminal voltage decrease. When the terminal voltage drops below a certain value, the phone will shut down.



An **ideal power source has no resistance**. The resistance of a dry cell is about 0.5 Ω and that of a 0–12 V power pack is about 2–3 Ω.

Example 8 Experiment on internal resistance of battery

Willy uses the circuit below (Fig a) to find the internal resistance r of a battery of e.m.f. ε. After closing the switch, he varies the resistance R of the variable resistor and records the corresponding voltmeter reading V and ammeter reading I. The V–I graph obtained is as shown (Fig b). Assume that the ammeter and the voltmeter are ideal.

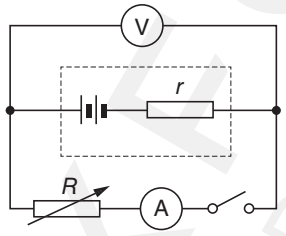


Fig a

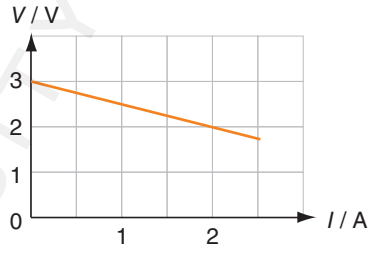


Fig b

(a) Express V in terms of ε, I and r.

(b) Find ε and r.

(c) State a precaution for this experiment.

(d) What is R when (i) $V = 2$ V; (ii) $V = 0$?

Solution

(a) $\varepsilon = V + Ir \Rightarrow V = (-r)I + \varepsilon$

Tip
$V = (-r)I + \varepsilon$
y-intercept of V–I graph = ε
Slope of V–I graph = $-r$

(b) $\varepsilon = 3$ V

$$r = -\text{slope} = -\frac{2-3}{2-0} = 0.5\ \Omega$$

(c) Set the resistance of the variable resistor to its maximum value before starting the experiment. OR

Open the switch immediately after taking each set of readings.

This prevents the current from being too large so that the circuit components will not be damaged.

This prevents the battery from heating up so that r remains constant.

(d) (i) From Figure b, $V = 2$ V $\Rightarrow I = 2$ A

$$R = \frac{V}{I} = \frac{2}{2} = 1\ \Omega$$

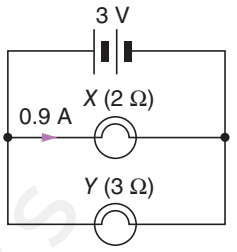
(ii) By $V = IR$, since $I \neq 0$, $V = 0 \Rightarrow R = 0$

The battery is shorted.

r is the only load and takes up all the voltage.

Example 9 Practical battery and ideal battery

In the circuit, the battery has an e.m.f. ε of 3 V and internal resistance r. The current through X is 0.9 A.



(a) Find the terminal voltage V.

(b) Find r.

(c) How will the current through bulb X change if bulb Y burns out?

(d) What will the answer to (c) be if the battery is ideal?

Solution

(a) $V = V_X = I_X R_X = 0.9 \times 2 = 1.8$ V

(b) $I_Y = \dfrac{V_X}{R_Y} = \dfrac{1.8}{3} = 0.6$ A

$V_Y = V_X$

$I = I_X + I_Y = 0.9 + 0.6 = 1.5$ A

By $\varepsilon = V + Ir$,

$3 = 1.8 + (1.5)r$

$r = 0.8\ \Omega$

Caution
Do not forget r.

(c) Total resistance R of circuit $= 2 + 0.8 = 2.8\ \Omega$

Current through $X = \dfrac{\varepsilon}{R} = \dfrac{3}{2.8} = 1.07$ A > 0.9 A

$\therefore$ The current increases.

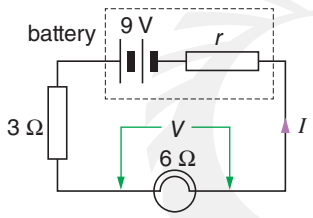
Tip
X and Y are connected in parallel to an ideal battery.
$V_X = \varepsilon$ and is not affected by the parallel branch containing Y.

(d) The current through X will remain unchanged.

Quick check 7

Google Forms Microsoft Forms

1



Complete the table.

	r / Ω	I / A	V / V
(a)	0		
(b)	1		
(c)			4.5

Learn by practice 2

1 A battery has an e.m.f. of 9 V. When it is connected to an 8-Ω resistor, the current through the resistor is 1 A. Find its internal resistance.

2 A battery has an e.m.f. of 3 V and internal resistance of 0.5 Ω. It is connected to a 4-Ω resistor. Find the voltage across the resistor.

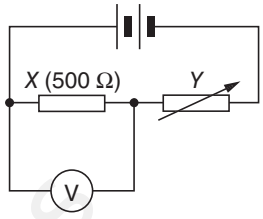
3 A battery has an e.m.f. of 6 V. When it is connected to a 10-Ω light bulb, its terminal voltage is 5 V. Find its internal resistance.

4 A variable resistor R is connected to a battery with internal resistance. Explain how the voltage across R changes when its resistance decreases.

E-version in OUP Exercise Platform is also available.

Writing practice

In the circuit shown, the internal resistance of the battery is 1 Ω. A voltmeter of resistance 1 kΩ is used to measure the voltage across *X*.



Question 1

Explain why the voltmeter reading is smaller than the theoretical value. State how a more accurate result can be obtained. (3 marks)

Answer

When the voltmeter is connected in parallel to *X*, their equivalent resistance becomes lower than 500 Ω. 1A

Therefore, the voltage across *X* becomes lower. 1A

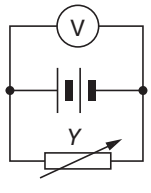
A more accurate result can be obtained by using a voltmeter of a higher resistance. 1A

Question 2

Describe an experiment to show how the terminal voltage of the battery is affected by the total resistance of the circuit. You may use the battery, the variable resistor *Y* and the voltmeter in the experiment. Draw a diagram for the circuit used and state the expected result. (3 marks)

Answer

(Correct circuit with correct symbol) 1A

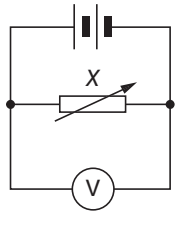


Vary the resistance of *Y* and record the corresponding voltmeter readings. 1A

When the total resistance of the circuit decreases, the terminal voltage decreases. 1A

Practise yourself

In the circuit shown, the internal resistance of the battery is *r*. A voltmeter of resistance 1 kΩ is used to measure the voltage across *X*.



1 Compare the voltmeter reading and the theoretical value. Explain. (3 marks)

2 Explain how the voltmeter reading changes when the resistance of *X* increases. (3 marks)

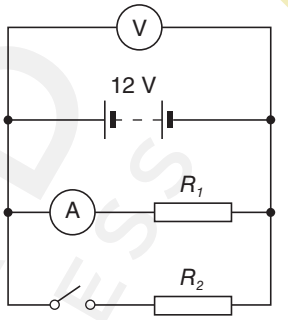
☆ HKDSE 2015 Paper 1A Q26

→ This video demonstrates how this question is solved.

Video

DSE practice 3 Parallel connection to a battery with internal resistance

In the circuit shown, the battery has internal resistance. The ammeter and the voltmeter are ideal. How will the readings of the ammeter and the voltmeter change when the switch is closed?



	Ammeter reading	Voltmeter reading
A	increases	increases
B	increases	decreases
C	decreases	increases
D	decreases	decreases

Solution

After closing the switch, the equivalent resistance of the external circuit decreases.

The p.d. shared by the external circuit, hence the p.d. across R_1, decreases.

As a result, both the readings of the ammeter and voltmeter decrease.

∴ The answer is D.

Common mistake ⚠
Students may wrongly think that the p.d. across the external circuit does not change when the switch is closed. This is true only if the battery is ideal.

Revision 3 Q12 (p.137)

Exercise 3.2

Unless otherwise specified, ammeters, voltmeters and batteries are regarded as ideal.

★1

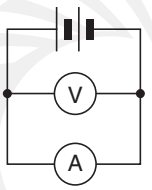


Fig a

A voltmeter and an ammeter are connected to a battery as shown (Fig a). Which of the following will happen?

(1) The voltmeter burns out.

(2) The ammeter burns out.

(3) The voltmeter reading is greater than 0.

A (1) only B (2) only

C (1) and (2) only D (2) and (3) only

★2

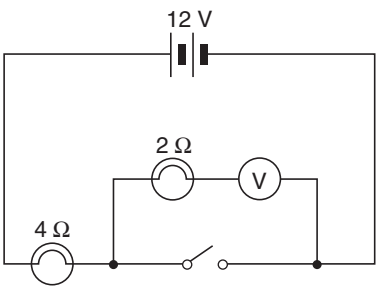


Fig b

Which of the following statements about the circuit above is/are correct?

(1) When the switch is open, no current flows through the 4-Ω bulb.

(2) When the switch is open, the voltmeter reading is 12 V.

(3) When the switch is closed, the voltmeter reading is 4 V.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

*3 A student uses the circuit in Figure c to measure the resistance of a resistor R. Both the voltmeter and the ammeter are **not** ideal.

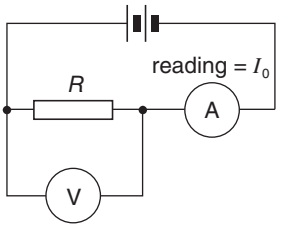


Fig c

Which of the following statements is/are correct?

(1) Replacing the voltmeter by an ideal one could obtain the actual voltage across R.

(2) Replacing the ammeter by an ideal one could obtain the actual current through R.

(3) $\frac{V_0}{I_0}$ is greater than the actual resistance of R.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

*4 What is the ammeter reading in the circuit (Fig d)?

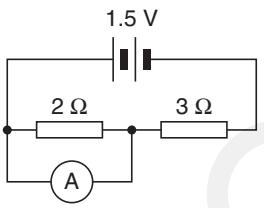


Fig d

*5 An ammeter of resistance 5 Ω and a voltmeter of resistance 10 kΩ are used to determine the resistance r of a resistor (Fig e).

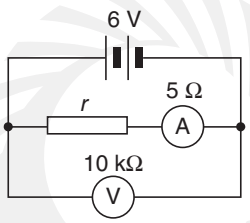


Fig e

(a) What does the ammeter measure?

(b) What does the voltmeter measure?

(c) Find the percentage error in the value of r obtained using this circuit if

(i) $r = 2\ \Omega$;

(ii) $r = 2\ \text{k}\Omega$

*6 In the circuits below (Fig f), all the batteries and all the resistors are identical. The batteries have non-zero internal resistance.

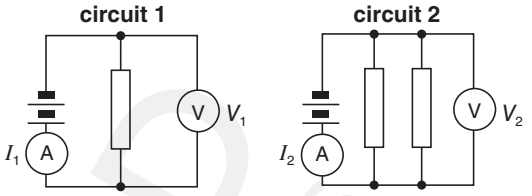


Fig f

(a) Which ammeter reading, I_1 or I_2, is larger? Explain your answer.

(b) Which voltmeter reading, V_1 or V_2, is larger? Explain your answer.

*7 In the circuit shown (Fig g), the battery has an e.m.f. of 6 V and internal resistance of 2 Ω. The resistance of the variable resistor R is varied and the corresponding voltmeter reading V and ammeter reading I are recorded.

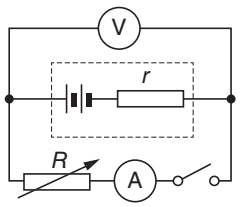


Fig g

(a) Sketch the graph of V against I.

(b) What are the physical meanings of the slope and y-intercept of the graph in (a)?

(c) If the battery is replaced by another one with a higher internal resistance, sketch the new graph in the graph in (a).

*8 In the circuit shown (Fig h), the voltmeter has a resistance of 1000 Ω.

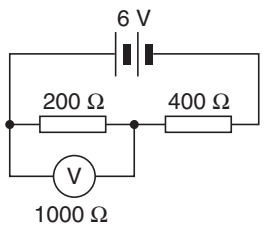


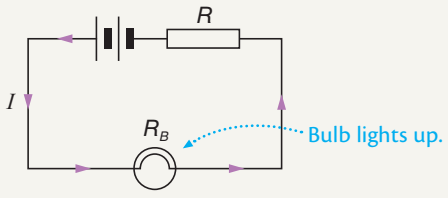
Fig h

(a) Find the voltage across each resistor.

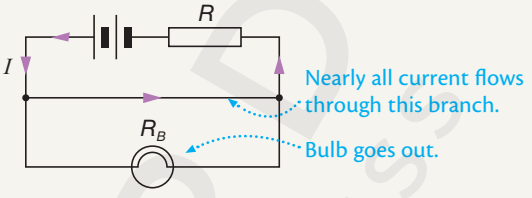
(b) If the voltmeter is then connected across the 400-Ω resistor, how will the voltage across each resistor change?

Key concepts 3

1 Short circuit



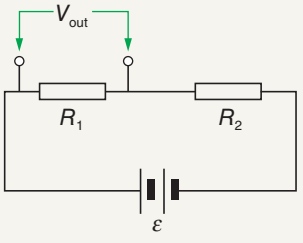
total resistance of circuit = $R + R_B$



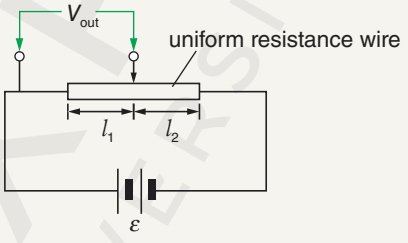
total resistance of circuit ≈ R

When a wire is connected across the bulb, a short circuit is formed, and the bulb goes out. The current through R increases.

2 Potential divider



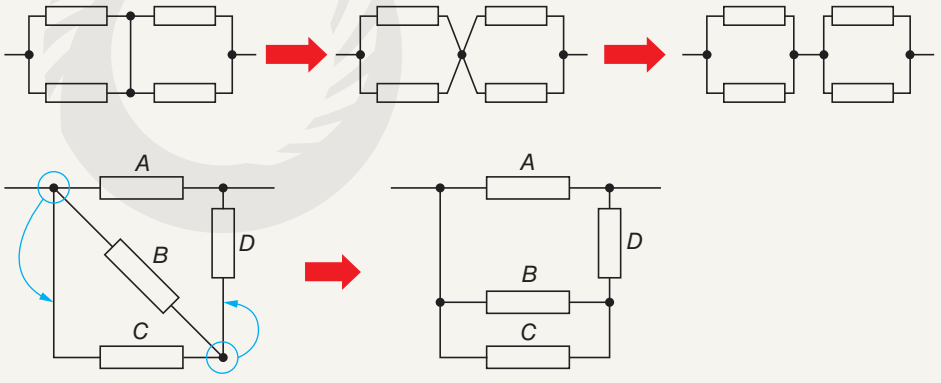
$$V_{out} = \frac{R_1}{R_1 + R_2} \times \varepsilon$$



$$V_{out} = \frac{l_1}{l_1 + l_2} \times \varepsilon$$

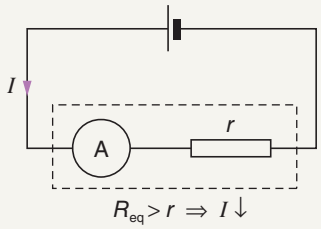
3 Resistor network

Rearrange the resistors to identify whether they are in series or parallel.

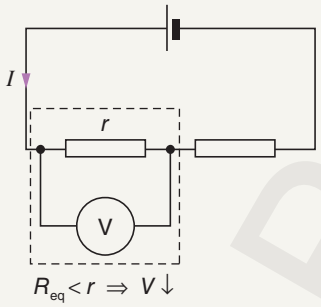


4 Internal resistance

- A good ammeter has a very low resistance.

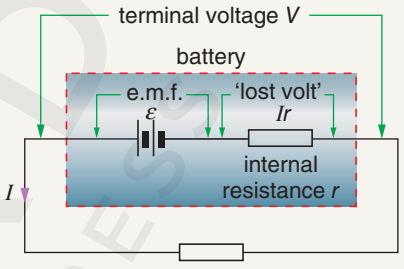


- A good voltmeter has a very high resistance.



- For a battery with an internal resistance r,

$$\varepsilon = V + Ir$$



Key terms

1	internal resistance 內電阻	p.120	**3**	short circuit 短路	p.112
2	potential divider 分壓器	p.114	**4**	terminal voltage 端電壓	p.128

Revision 3

Unless otherwise specified, ammeters, voltmeters and batteries are regarded as ideal.

★1 3.1 In the circuit below (Fig a), when switch S is opened, the resistance between X and Y is 2 Ω.

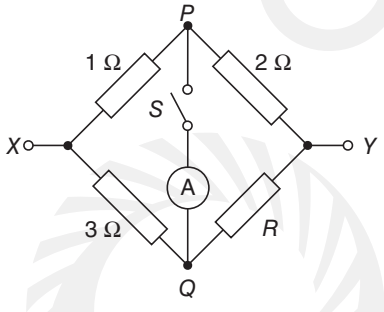


Fig a

Which of the following is/are correct?

(1) $R = 3\ \Omega$

(2) When switch S is closed, the resistance between X and Y is reduced.

(3) When switch S is closed, the ammeter records a non-zero reading.

A (1) only B (3) only
C (1) and (3) only D (1), (2) and (3)

★2 3.2

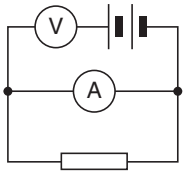


Fig b

What will happen to the ammeter and voltmeter in the above circuit (Fig b)?

	Ammeter	Voltmeter
A	burns out	burns out
B	burns out	reading > 0
C	reading = 0	burns out
D	reading = 0	reading > 0

★3 3.1 In the circuit below, all the bulbs are identical (Fig c).

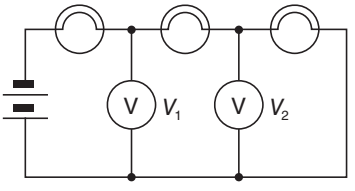


Fig c

If the reading of voltmeter V_1 is 3 V, what is the reading of voltmeter V_2?

A 1 V B 1.5 V
C 3 V D 6 V

★4 3.2 What are the readings of the ammeter and the voltmeter (Fig d)?

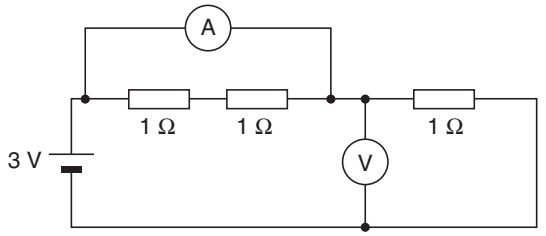


Fig d

	Ammeter	Voltmeter
A	zero	1 V
B	1 A	1 V
C	3 A	1 V
D	3 A	3 V

★5 3.1 In the resistor network below (Fig e), each resistor has resistance of 1 Ω.

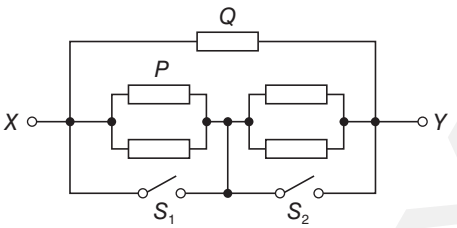


Fig e

Which of the following configurations has the highest resistance between point X and Y?

A S_1 open, S_2 open

B S_1 open, S_2 closed

C S_1 closed, S_2 open

D S_1 closed, S_2 closed

★6 3.1 In the circuit shown (Fig f), how do the ammeter reading and voltmeter reading change when the switch is closed?

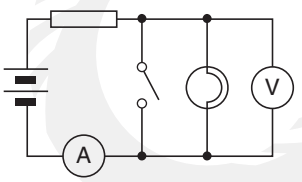


Fig f

	Ammeter reading	Voltmeter reading
A	increases	becomes zero
B	increases	remains unchanged
C	remains unchanged	becomes zero
D	remains unchanged	remains unchanged

★★7 3.1 In the circuit below (Fig g), XY is a uniform wire of resistance 20 Ω. Contact point S can slide along XY. The resistance of bulb Q is 1 Ω. The e.m.f. of the battery is 6 V.

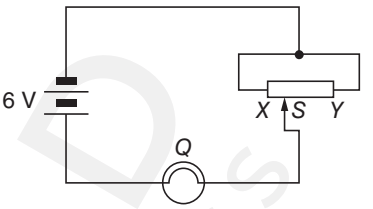


Fig g

Which of the following statements are correct?

(1) Bulb Q is the dimmest when S is at the midpoint of XY.

(2) The current drawn from the battery is 1 A when S is at the midpoint of XY.

(3) The adjustable range of p.d. across Q is 1–6 V.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

▶ Refer p.114, 117

★★8 3.1 Four identical bulbs P, Q, R and S are connected to an ideal battery (Fig h).

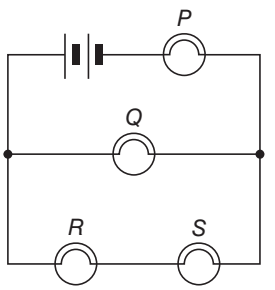


Fig h

The power consumed by bulb Q is 1 W. Which of the following statements is/are correct?

(1) The power consumed by bulb P is 2.25 W.

(2) The power consumed by bulb R is 0.5 W.

(3) If bulb S is shorted, bulb Q will become dimmer.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

▶ Refer p.113, 117

9 HKDSE 2013 Paper 1A Q31
3.1

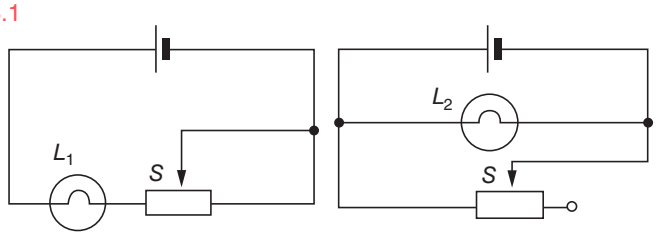


Fig i

In each of the above circuits (Fig i), the cell has constant e.m.f. and negligible internal resistance. When the sliding contact S of each rheostat shifts from the mid-position to the right, how would the brightness of each bulb change?

	Bulb L_1	Bulb L_2
A	becomes dimmer	remains unchanged (43%)
B	becomes dimmer	becomes brighter (29%)
C	remains unchanged	becomes dimmer (17%)
D	becomes brighter	remains unchanged (11%)

10 HKDSE 2013 Paper 1A Q32
3.1

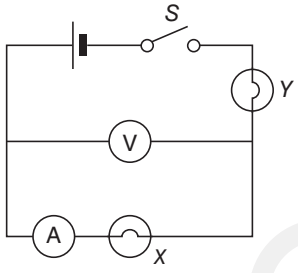


Fig j

Exam Report: It seems that more that half of the candidates did not fully understand the meaning of short circuit and open circuit.

In the above circuit (Fig j), the cell has negligible internal resistance. When switch S is closed, both bulbs are not lit. The voltmeter has a reading but the ammeter reads zero. If only one fault has been developed in the circuit, which of the following is possible?

A Bulb X has been shorted accidentally. (15%)
B Bulb Y has been shorted accidentally. (15%)
C Bulb X is burnt out and becomes open circuit. (46%)
D Bulb Y is burnt out and becomes open circuit. (24%)

11 HKDSE 2014 Paper 1A Q24
3.2

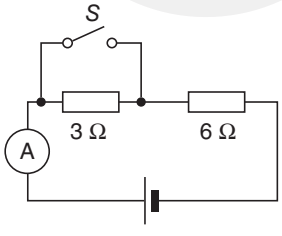


Fig k

Exam Report: It seems that over 60% of the candidates simply neglected the effects of the cell's internal resistance.

In the above circuit (Fig k), the cell has constant e.m.f. and a fixed internal resistance. When S is closed, the ammeter reads 3.0 A. When S is open, which of the following is a possible reading of the ammeter?

A 1.6 A (9%)
B 2.0 A (63%)
C 2.4 A (20%)
D 3.2 A (8%)

12 HKDSE 2015 Paper 1A Q26
3.2

In the following circuit, the cell has a finite internal resistance and both meters are ideal.

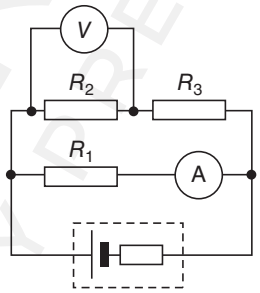


Fig l

In which situation below will the readings of the ammeter and the voltmeter suddenly increase?

A R_1 is faulty and becomes a short circuit.
B R_2 is faulty and becomes a short circuit.
C R_3 is faulty and becomes a short circuit.
D R_2 is faulty and becomes an open circuit. (26%)

13 HKDSE 2020 Paper 1A Q23
3.2

Three identical resistors, a battery of negligible internal resistance and an ideal voltmeter are connected to form circuits (a) and (b) respectively (Fig m).

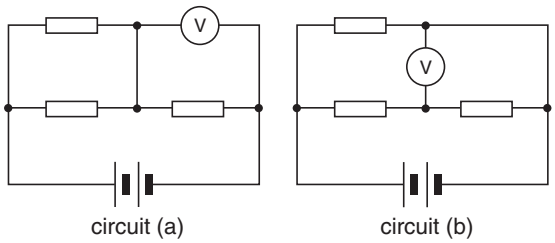


Fig m

Given that the voltmeter reading is 8 V in circuit (a), what is the voltmeter reading in circuit (b)?

A 4 V (25%)
B 6 V (32%)
C 8 V (24%)
D 12 V (19%)

Conventional questions

***14** 3.2 Three resistors, 3 Ω, 4 Ω and 12 Ω, are connected to a 6-V battery (Fig n).

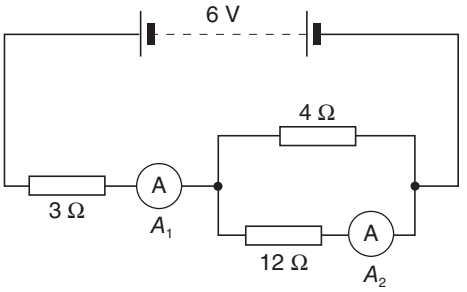


Fig n

(a) Find the readings of A_1 and A_2. (3 marks)

(b) If the 6-V battery is replaced by another battery of unknown e.m.f., the reading of A_1 will be 0.4 A. Find the reading of A_2 (2 marks)

(c) If A_2 is replaced by a voltmeter, find

(i) the reading of A_1, (1 mark)

(ii) the reading of the voltmeter. (1 mark)

(d) If A_1 is replaced by a voltmeter, find

(i) the reading of A_2, (1 mark)

(ii) the reading of the voltmeter. (1 mark)

***15** 3.1 The three resistors and the bulb in the circuit below have the same resistance r (Fig o). The resistance R of the variable resistor is adjustable between 5–60 Ω.

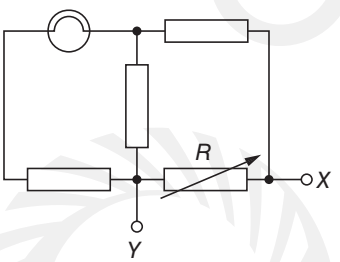


Fig o

(a) Express the resistance between X and Y in terms of r and R. (2 marks)

(b) The resistance between X and Y is measured to be 10 Ω when R is set to be 20 Ω. Find r. (2 marks)

(c) What is the resistance adjustable between X and Y? (2 marks)

(d) A constant p.d. is applied across X and Y. How does the brightness of the bulb change when R increases? (1 mark)

***16** 3.2 A student is given a resistor, a battery, a voltmeter with internal resistance 1000 Ω and an ammeter with internal resistance 1 Ω. He finds that he can use two different circuits to measure the unknown resistance R of the resistor (Fig p).

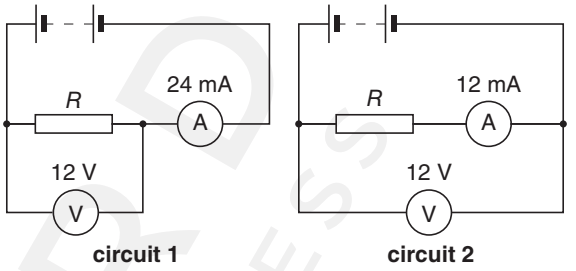


Fig p

(a) Estimate the value of R using the readings of the voltmeter and the ammeter in each circuit. (2 marks)

(b) Considering the value of R found in (a), and the resistance of the ammeter and voltmeter, which circuit gives a more accurate estimation of R? Explain your answer. (4 marks)

***17** Misc In the circuit shown in Figure q, XY is a uniform resistance wire of resistance 10 Ω. The e.m.f. of the battery is 6 V. The battery has an internal resistance of 1 Ω.

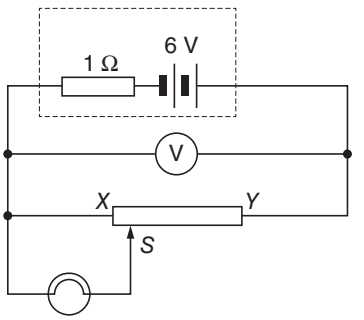


Fig q

(a) (i) Describe the change in the brightness of the bulb as the sliding contact S moves from X to Y. (1 mark)

(ii) Describe the change in voltmeter reading as the sliding contact S moves from X to Y. (1 mark)

(b) The current drawn from the battery is 1 A when S is at Y.

(i) Find the resistance of the bulb. (2 marks)

(ii) Find the terminal voltage of the battery. (2 marks)

★18 3.2 A camera is powered by a 3.6-V rechargeable battery as shown (Fig r). At 25 °C, the camera draws an average current of 1 A in taking each picture. The internal resistance of the battery rises as the temperature drops (Fig s). Assume the resistance of the camera remains constant.



Fig r

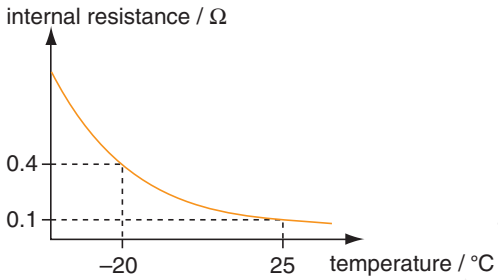


Fig s

(a) Estimate the resistance of the camera when taking a picture at 25 °C. (3 marks)

(b) Estimate the current drawn by the camera when taking a picture at –20 °C. (2 marks)

(c) Estimate the terminal voltage of the battery at –20 °C. (2 marks)

(d) Explain why the camera may not work properly at a low temperature (e.g. –20 °C). (1 mark)

★19 3.2 Three resistors, 1 kΩ, 10 kΩ and 10 kΩ, are connected in series to a 6-V battery of negligible internal resistance (Fig t).

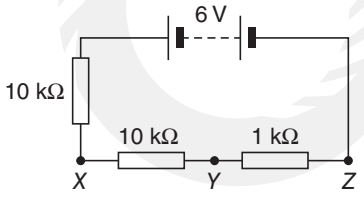


Fig t

(a) What is the p.d. across

(i) X and Y?

(ii) Y and Z? (2 marks)

(b) A voltmeter of resistance R_V = 10 kΩ is used to measure the potential difference across the resistors. What is the voltmeter reading when it is connected across

(i) X and Y?

(ii) Y and Z? (4 marks)

(c) By comparing the results in (a) and (b), across which two points, X and Y or Y and Z, does the voltmeter give a relatively inaccurate value? Hence state the general principle of selecting a suitable voltmeter for such measurements. (3 marks)

★20 Misc A high voltage probe (Fig u) is used for measuring voltage up to 40 kV.

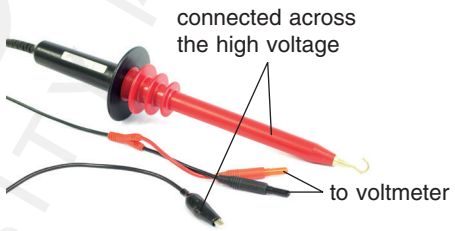


Fig u

The probe is basically a potential divider (Fig v). It allows a voltmeter of range 0–5 V to measure the high voltage.

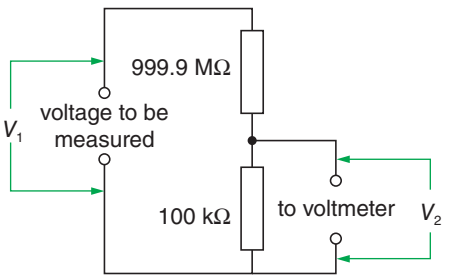


Fig v

The voltage to be measured is V_1 and the reading of the voltmeter is V_2.

(a) Resistors of very high resistance are used in the circuit. Explain an advantage of doing so. (2 marks)

(b) Assume the voltmeter is ideal. What is the ratio of $V_1 : V_2$? (1 mark)

(c) If the voltmeter has a resistance of 100 kΩ and V_2 is 4 V, what is V_1?
Hence find $V_1 : V_2$. (3 marks)

★21 Misc When the chemical energy stored in battery B is consumed, the internal resistance r and the e.m.f. of B change (Fig w and x).

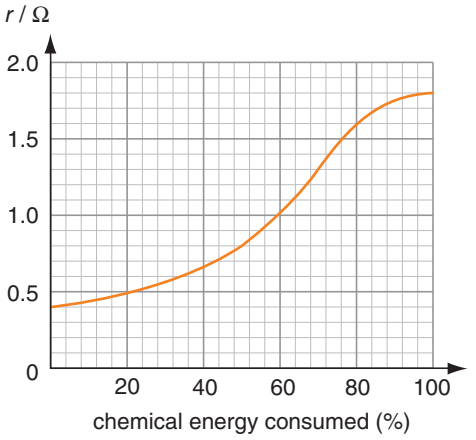


Fig w

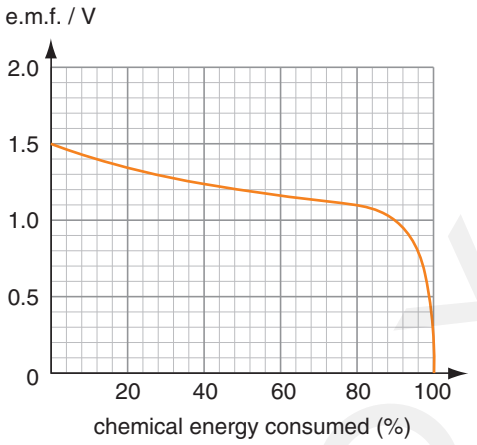


Fig x

(a) B is connected to a 5-Ω light bulb. What is the terminal voltage of B when it is

(i) new? (2 marks)

(ii) 50% consumed? (1 mark)

(b) Figure y shows a simplified structure of a battery tester.

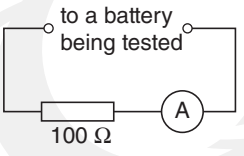


Fig y

(i) B is connected to the tester. Find the current measured by the ammeter when B is

(1) 50% consumed. (1 mark)

(2) 80% consumed. (1 mark)

(ii) Suggest the function of the 100-Ω resistor. (1 mark)

★22 3.1 P and Q are two identical bulbs (Fig z). XY is a uniform wire of resistance 3 Ω. Initially, the sliding contact S is in the middle between XY.

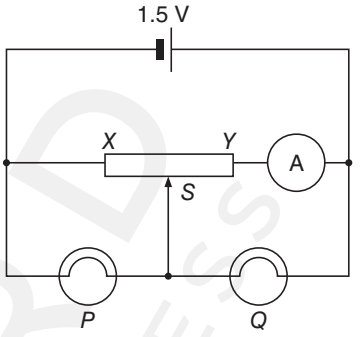


Fig z

(a) Describe the change in brightness of bulbs P and Q if

(i) S is moved towards X; (2 marks)

(ii) S is at X. (2 marks)

(b) The ammeter reading is 2 A when S is at Y. Find the resistance of P. (3 marks)

(c) Where is S when the power delivered by the battery is the maximum? (3 marks)

★23 3.2 Michelle studies the internal resistance of dry cell X using the circuit below (Fig aa). She varies the resistance of the variable resistor and plots a graph of the voltmeter reading V against the ammeter reading I (Fig ab).

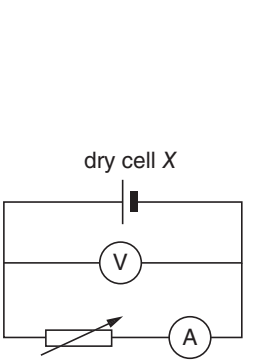


Fig aa

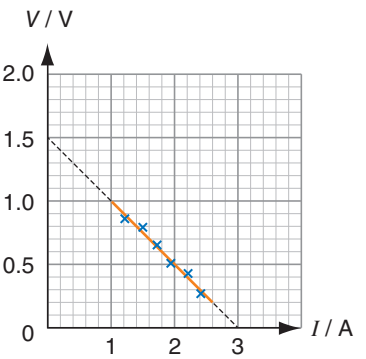


Fig ab

(a) Express V in terms of I, ε (the e.m.f. of the cell) and r (the internal resistance of the cell). (1 mark)

(b) Deduce from the graph the e.m.f. and internal resistance of the cell. (2 marks)

(c) Cell X is replaced by cell Y which has the same e.m.f. but higher internal resistance than cell X. Draw a line on Figure ab to show the results of the experiment for Y. (2 marks)

★24 3.1 The resistances of A, B, C and D are $3R$, R, $2R$ and $4R$ respectively and they are connected to a battery as shown (Fig ac).

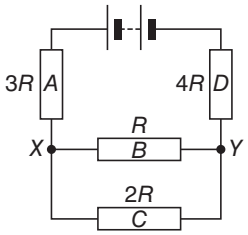


Fig ac

The power dissipation of B is 4 W.

(a) Find the power dissipation of C. (2 marks)

(b) Find the equivalent resistance of B and C. (1 mark)

(c) Find the power dissipation of A. (2 marks)

(d) Find the output power of the battery. (2 marks)

(e) How will the answer to (d) change if B is shorted? (2 marks)

★25 3.1 A strand of Christmas decoration lighting (Fig ad) consists of 25 identical filament bulbs connected in series. Each bulb has a shunt wire connected in parallel to the filament (Fig ae). The p.d. across the strand is 50 V; while the current through the bulbs is 0.25 A.



Fig ad

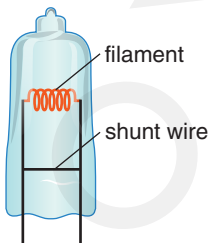


Fig ae

(a) Find the resistance of each bulb. (2 marks)

(b) The shunt wire is covered with an oxide coating. When a bulb works normally, the resistance of its shunt wire is 20 times that of its filament. When the filament of a bulb breaks, the current flowing through the shunt wire will burn off the coating and reduce the shunt's resistance to 2 Ω.

(i) Find the power consumed by a shunt wire when all of the bulbs are working normally. (4 marks)

(ii) If the filament of a bulb in a strand burns out, what is the power consumed by the shunt wire in that bulb before the coating is burnt off? (3 marks)

(iii) Explain the function of the shunt wire. (2 marks)

(iv) If some of the bulbs in the strand burn out, would the remaining bulbs become brighter or dimmer? (1 mark)

★★26 Misc In a STEM project, Mary makes a meter to measure stretching forces (Fig af). XY is a uniform resistance wire of length 30 mm and resistance 30 Ω. P is a sliding contact. It touches X when no stretching force is applied on the meter. When the meter is stretched at both ends, P will slide smoothly towards Y. The voltage of the battery is 6 V and the resistance of the variable resistor is set to 70 Ω. The voltmeter reading then indicates the length XP, which is related to the force stretching the meter as shown (Fig ag).

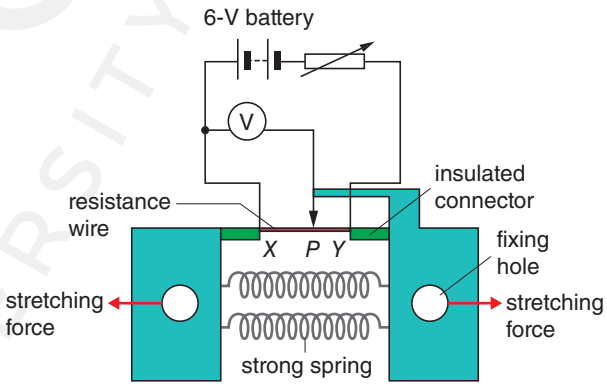


Fig af

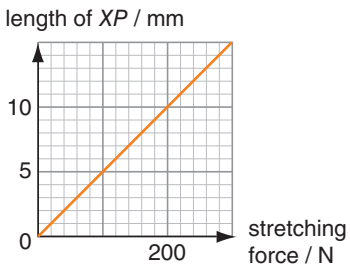


Fig ag

(a) Show that the voltmeter reads 1.8 V when P touches end Y of the wire. (2 marks)

(b) Find the stretching force when the voltmeter reading is 0.6 V. (2 marks)

(c) What is the function of the variable resistor? What will happen if its resistance increases? (2 marks)

Refer p.114

See p.143 for the Exam Report.

27 HKDSE 2013 Paper 1B Q10

Misc

(a) In the circuit shown in Figure ah, a 12 V battery of negligible internal resistance is connected with a thermistor R and a resistor of resistance 120 Ω. The graph shows the variation of the thermistor's resistance with temperature.

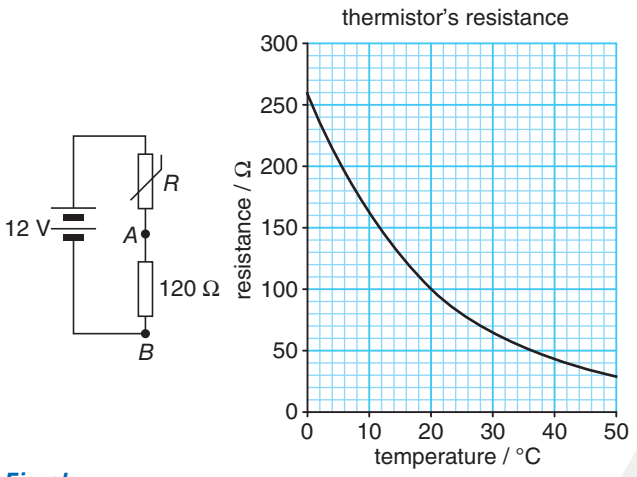


Fig ah

(i) Find the resistance of the thermistor R at 25 °C. (1 mark)

(ii) What is the potential difference V_{AB} across A and B at 25 °C? (2 marks)

(b)

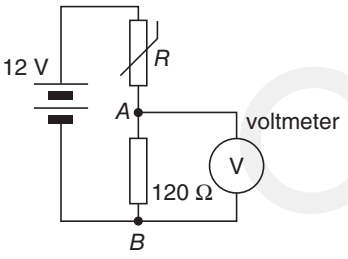


Fig ai

Kelly wants to confirm the above calculation by measuring V_{AB} using a voltmeter of about 1 kΩ resistance (Fig ai). She finds that the reading registered is slightly different from the value found in (a) despite making careful measurements. Explain why this is so. Suggest how the accuracy of the measurement could be improved. (3 marks)

(c) (i)

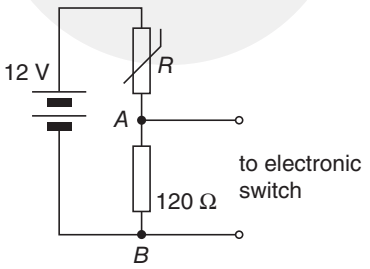


Fig aj

The potential difference V_{AB} is used to drive an electronic switch connected across AB to turn on a fan if temperature rises above a certain value such that V_{AB} is 6.0 V or above (Fig aj). Using the information provided in Figure ah, find the minimum temperature needed to keep the fan on. Show your working. (2 marks)

(ii)

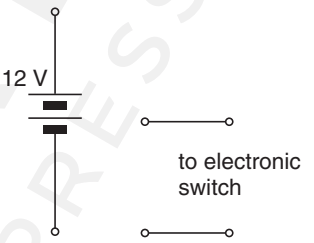


Fig ak

Without using additional components, complete the new circuit diagram in Figure ak to illustrate how the circuit can be modified to turn on a heating device when temperature falls in Figure ak a certain value. Explain the action of the circuit. No calculation is required. (3 marks)

See p.143 for the Exam Report.

28 HKDSE 2016 Paper 1B Q7

(a) The circuits in Figure al each contains two resistors connected in series with a 6 V battery of negligible internal resistance. The resistors in circuit I are 10 kΩ each while those in circuit II are 100 Ω each.

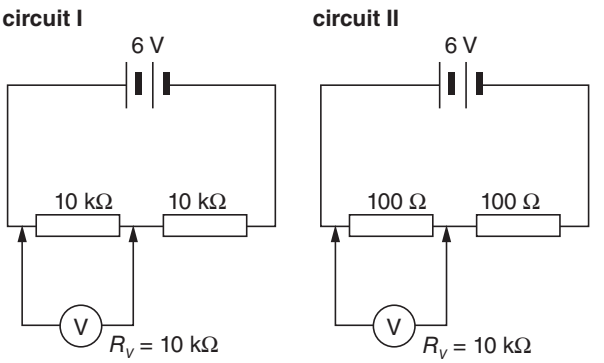


Fig al

A voltmeter of internal resistance R_V = 10 kΩ is used to measure the potential difference across one of the resistors as shown.

(i) What would be the respective voltmeter readings? (3 marks)

(ii) In fact, the potential difference across each resistor *before connecting the voltmeter* is 3 V in both circuits. Explain why this voltmeter gives a relatively inaccurate value for circuit I. Hence state the general principle of selecting a suitable voltmeter for such measurement. (2 marks)

(b) Circuit III shows a possible method for measuring resistance using a voltmeter and an ammeter (Fig am). The internal resistances of the voltmeter and the ammeter are R_V and R_A respectively and their readings V_m and I_m give the measured $R_m = \frac{V_m}{I_m}$. The true resistance value of the resistor is R.

Circuit III

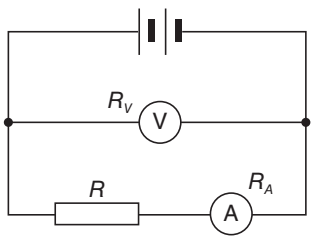


Fig am

(i) State which reading(s), V_m, I_m or both, do(es) **NOT** give the *true voltage* across the resistor and/or the *true current* passing through the resistor. Hence write down an equation relating R_A, R_m and R. (2 marks)

(ii) Find the percentage error associated with R_m when measuring the resistance of this resistor.

Given: $R_V = 10\ k\Omega$, $R_A = 1\ \Omega$ and $R = 10\ \Omega$ (2 marks)

Q27 Exam Report: Part (a) was in general well answered. Many candidates attempted to tackle part (b) in terms of current passing through the voltmeter rather than the voltage across *AB*. As a result, many failed to explain precisely how the reading was affected. In (c)(i), for those who were able to find the resistance of *R* by considering the potential difference across the resistors, most managed to work out the correct answer. Some candidates did not attempt part (c)(ii) which involved unfamiliar situation. For those who did attempt the question, not many were able to explain the action of the circuit.

29 HKDSE 2019 Paper 1B Q7

You are provided with a battery (of fixed e.m.f. ε and internal resistance r), a variable resistor (with several known resistance values R to be selected), a switch, a voltmeter (assumed ideal) and a few connecting wires.



Fig an

(a) With the aid of a circuit diagram, describe the procedure of an experiment to study how the terminal voltage V delivered by the battery depends on the resistance R connected to it. State **one** precaution of the experiment. (5 marks)

(b) Describe the variation of V with R and express V in terms of ε, r and R. (2 marks)

Q28 Exam Report: This question tested candidates' understanding of the loading effect of meters in simple circuits. Candidates' performance in (a) was satisfactory. In (a)(i), some candidates took the product of the total current and the resistance of the resistor concerned as the voltmeter reading. Most candidates knew that the finite internal resistance of the voltmeter introduced error to the measurement but few understood how to work out the problem mathematically. Some candidates in (a)(ii) failed to state explicitly that the resistance of the voltmeter should be 'much higher' than that of the part under measurement. In (b)(i), many candidates wrongly held that both voltmeter and ammeter readings are not true values for the resistor concerned. Not many obtained the correct percentage error in *R* in (b)(ii).

Q29 Exam Report: Part (a) was in general well answered. Most candidates were able to draw the circuit diagram correctly though some failed to mark the polarities of the voltmeter or wrongly connected it across the switch. Quite a number of the candidates suggested precautions that were not relevant to accuracy or safety of the experiment like 'using a voltmeter (assume idea) of a very high resistance'. In (b), not many gave a correct expression of *V* in terms of ε, *r* and *R*. A significant number of them mistook that *V* varied linearly with *R* or they had a directly proportional relationship.

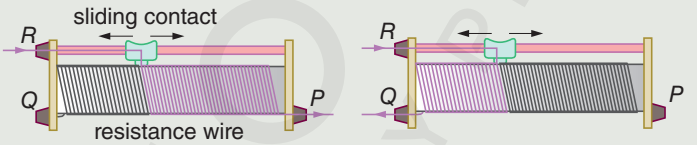
Know more

Rheostat as potential divider (p.114)

A rheostat (Fig a) can be used as a variable resistor when two terminals are used (Fig b). It can also be used as a potential divider when all the 3 terminals are used (Fig c). V_{out} is the voltage across *C* and *P*. When the sliding contact is moved towards *Q*, V_{out}, thus the voltage across the light bulb, increases. This makes the bulb brighter.



Fig a



(i) connected across P and R *(ii) connected across Q and R*

Fig b

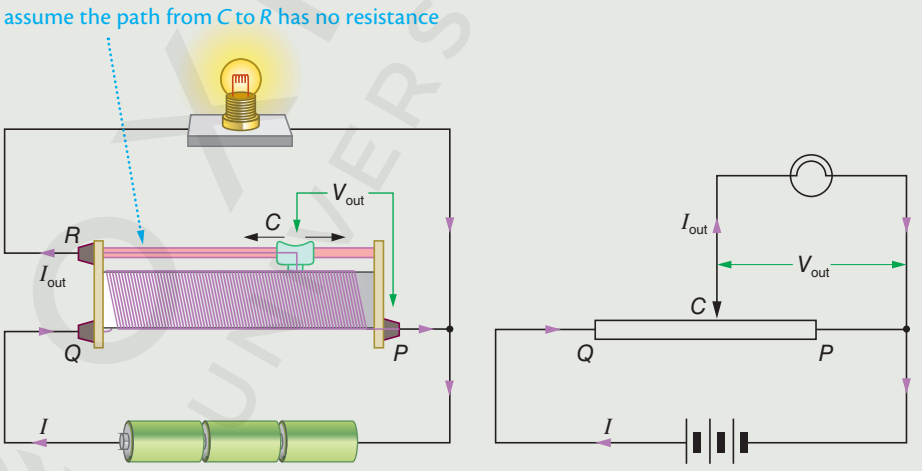


Fig c